



# The effects of vehicular emissions on the activity and diversity of the roadside soil microbial community<sup>☆</sup>

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## ABSTRACT

Motor vehicles emit a variety of pollutants including metals, petroleum hydrocarbons and polycyclic aromatic hydrocarbons (PAHs). The relationships between metals, petroleum hydrocarbons and PAHs, soil respiration and microbial diversity (fungi and bacteria) were studied using control ( $n = 3$ ) and roadside soils ( $n = 27$ ) with different exposure periods to vehicle emissions (2–63 years). Bacterial diversity was found to be higher than control sites ( $P = 0.002$ ) but was the same across different categories of road age ( $P = 0.328$ ). Significant ( $r = -0.49$ ,  $P = 0.007$ ) contrasting behaviour of fungal and bacterial diversity was reported, with diversity increasing across all road types for bacteria and decreasing across all road types for fungi compared to control soils. Analysis of the bacterial community identified three distinct clusters, separated on age of contamination, suggesting that roadside bacterial communities change over time with pollution from vehicles with the potential development of metal resistant bacteria in roadside soils. In contrast, for fungal communities, a reduction in diversity with time of exposure to roadside vehicle emissions was observed suggesting the potential for reduced ecosystem functionality and soil health in roadside soils. This is the first study in the published literature to include both bacterial and fungal responses from aged roadside soils. The results from this study suggest that normal functionality of soil ecosystem services is being affected in roadside soils, potentially globally.

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## 1. Introduction

Given the large size of the global road network, the area of soil affected by vehicular emissions is extensive. As the estimated area for roadside soil in Australia itself is large, 80 million km<sup>2</sup>, and in the world, far more, these soils are likely to be disturbed in terms of biogeochemical processes which may affect the normal functionality of soil ecosystem services. The contaminants arising from vehicular emissions include metals,<sup>1</sup> petroleum hydrocarbons and polycyclic aromatic hydrocarbons (PAHs), all of which have the

potential to cause adverse effects on the diversity and activity of the roadside soil microbial community (Bååth, 1989; Elsgaard et al., 2001; Pérez-de-Mora et al., 2006; Renella et al., 2002).

Bacteria and fungi constitute more than 90% of the total soil microbial biomass and are key regulators of soil nutrient availability (Six et al., 2006). Bacteria and fungi tend to exhibit different responses to metal pollution (Rajapaksha et al., 2004), which could potentially lead to differences in the distribution, diversity and activity of microbial communities within roadside ecosystems. In general, soil microbial communities have been reported to recover following an initial high metal input (Holtan-Hartwig et al., 2002) and some are able to survive due to ecological resilience, resulting in a community showing increased resistance to metals over time (Griffiths and Philippot, 2013; Guzzo and DuBow, 1994).

Given the importance of soil microorganisms for ecosystem functioning, it is vital to understand how the soil microbial community responds to the presence of contaminants in roadside soils. The resilience of soil microbial communities is often claimed to be

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<sup>1</sup> Throughout this study use of the term 'metal' includes metals [such as Ag (silver), Cd (cadmium), Cu (copper), Cr (chromium), Mn (manganese), Mo (molybdenum), Ni (nickel), Pb (lead), Zn (zinc) and W (tungsten)], platinum group elements (PGE) [such as Pt (platinum), Rh (rhodium) and Pd (palladium)] as well as metalloids [such as As (arsenic) Sb (antimony) and Se (selenium)].

due to functional redundancy i.e. that the functions of contaminant-sensitive species can be replaced in terms of dominance within the community by contaminant-tolerant species that were already present in the soil in low numbers (Allison and Martiny, 2008). Microorganisms are also able to survive a wide array of environmental changes by gene alterations, leading to increased contaminant resistance (Guzzo and DuBow, 1994).

The co-occurrence of metals with PAHs is very common in roadside soil environments. Many studies have reported a significant reduction in microbial diversity and activity after both long and short term exposure to soil contaminated with metals (Bååth, 1989; Kandeler et al., 1996; Lorenz et al., 2006; Vivas et al., 2008). Like the effect of metals on soil microbes, petroleum hydrocarbons in the soil environment can also interfere with the soil microbiome (Borowik et al., 2017). Petroleum hydrocarbons in soil are known to be toxic to some soil microorganisms, resulting in changed soil microbial activity (Joner et al., 2001; Vivas et al., 2008) and diversity; however, at the same time this leads to selective pressure for particular species to thrive with increased degradation ability against PAH (Thavamani et al., 2012; Ameen et al., 2016; Marchand et al., 2017). Consequently, PAH-degrading bacteria have previously been isolated from roadside soil (Li et al., 2015; Tay et al., 2015).

To date, there have been few reports of the activity and diversity of the soil microbial community in roadside soils. Of the few publications available, most have focused on the effects of vehicle emissions on the soil bacterial community from just a few metals, such as Cd, Cu, Pb and Zn (Hao et al., 2009; Mohamed and Abo-Amer, 2012; Wang et al., 2011). Very few publications have focused on the effects of other metals known to be in vehicle emissions including Cr, Ni and Mn (Gülser and Erdoğan, 2008; Post and Beeby, 1996); however it has been shown that roadside soil-accumulated metals can affect microbial respiration (Brookes, 1995; Gülser and Erdoğan, 2008) and microbial biomass (Adelasoye and Ojo, 2014; Hao et al., 2009; Post and Beeby, 1996). One study reported that with increasing distance from the road, an increase in microbial activity was observed (Gülser and Erdoğan, 2008); other reports have shown either negative correlations with soil respiration (Gülser and Erdoğan, 2008) or no significant correlation with increasing distance from the road (Post and Beeby, 1996). This lack of consistency among publications may be due to variations in both the contaminants and their concentrations or to differences in soil characteristics and the associated microbial community. Roadside soil-PAH effects have been mainly reported in terms of changes in the bacterial community (Li et al., 2015; Tay et al., 2015) rather than the fungal community. Researchers have looked at bacterial and fungal communities in roadside soils separately but never in one study. They have also not focused on road age effects on the roadside microbial community. It remains unclear whether the soil microbial community i.e. bacteria and fungi exhibit the same or different responses to roadside contaminants (metals and PAHs), particularly for long-term exposure.

In the present study, roadside soil exposed to vehicular emitted metals over a range of times (from 1.5 to 63 years) were collected to study the effects of vehicular emitted metals and PAHs on soil microbial community. We hypothesized that as the age of the road increased, and therefore concentrations of vehicle emitted metals and PAHs in roadside soil, that there would be different impacts on roadside microbial communities. Total metals, PAHs and total petroleum hydrocarbons (TPH) in soil and CaCl<sub>2</sub>-extractable metals were measured along with soil physiochemical properties, soil nutrient status, and microbial biomass/respiration. Microbial analyses were performed to reveal the changes in both the activity and diversity of the bacterial and fungal community using quantitative polymerase chain reaction (qPCR) and denaturing gradient gel electrophoresis (DGGE).

## 2. Methods

### 2.1. Sample site selection

A total of 30 soil samples (Fig. 1) representing roadside soils ( $n = 27$ ) over a range of times of exposure to vehicle emissions (1.5 years–63 years) and non-roadside soil representing controls ( $>500$  m from the nearest road,  $n = 3$ ), with all sample sites representing high traffic ( $\geq 15,000$  vehicles/day) were collected from the west of Melbourne, Australia in December 2012 from soils developed on Newer Volcanic (basalt) geology (Price et al., 1997). One underlying geology ensured that the soil was derived from similar parent material to reduce variation in soil characteristics. All roads had the same surface material i.e. asphalt (Woodall and Thakur, 2013). Roadside soils were categorized as coming from new (1.5–3 years,  $n = 9$ ), medium (5–12 years,  $n = 9$ ) and old ( $\geq 15$  years,  $n = 9$ ) aged roads. Road ages, for individual roads were determined using VIC Road Government department data and library collections, State Library of Victoria repository collections and Greater Melbourne street directories editions 1966 to 2012. Control sites in parklands were chosen at locations  $>0.5$  km away from roads and  $>1$  km from industry. Topsoil (0–10 cm) was collected within a 2–5 m distance from road edges using a clean stainless-steel spatula and placed in zip-lock polythene bags (approximately 500 g). The immediate road edge up to 1 m was avoided to minimize the risk of sampling refilled or recently disturbed soil from road constructions and upgrades (Werkenthin et al., 2014). Samples were air dried and sieved using a 2 mm sieve. These samples were sealed in zip-lock polythene bags and stored at ambient laboratory conditions ( $22 \pm 2$  °C) in the dark until analyzed for metals, TPH, PAH and other soil parameters.

In the field a subset of fresh soil samples were transferred to 75 ml vials (approximately 10 g), and immediately stored in a cool box (Hjortenkrans et al., 2008; Werkenthin et al., 2014) for the determination of soil microbial activity. In the laboratory, the samples in the vials were transferred to a freezer ( $-10$  °C) until further analysis.

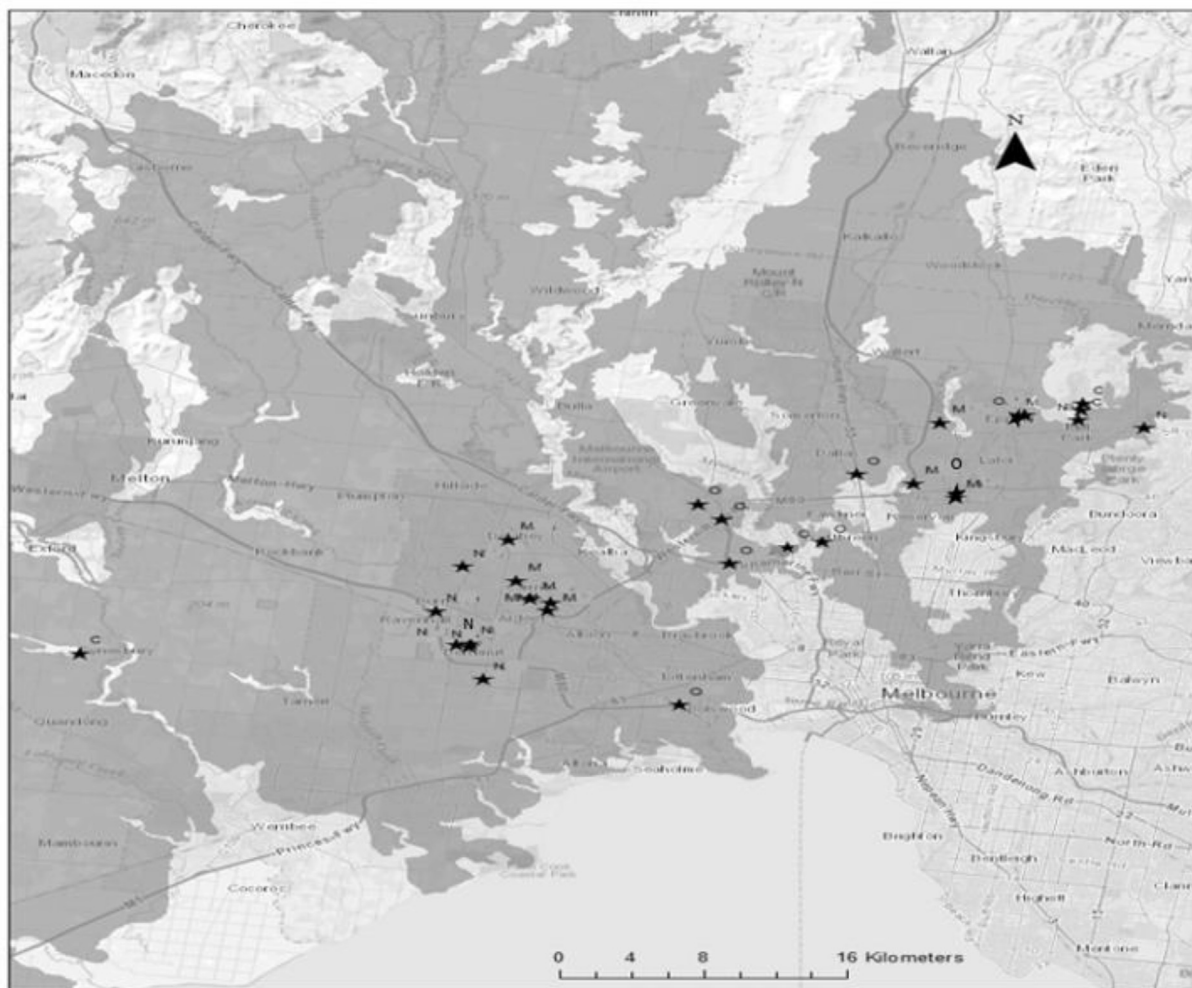
### 2.2. Physical and chemical properties of soil

Soil samples were analyzed for pH, effective cation exchange capacity (ECEC) and total organic carbon (TOC) as previously described (De Silva et al., 2016). Potassium chloride-(KCl)-extractable NO<sub>2</sub><sup>-</sup> and NO<sub>3</sub><sup>-</sup> was determined using the Cd reduction method (Griffin et al., 1995; Keeney and Nelson, 1983). Bicarbonate-extractable Colwell P was performed using continuous segment flow analysis (Colwell, 1963). Potassium chloride-extractable sulfur was determined by inductively coupled plasma atomic emission spectroscopy (ICPAES) and total organic carbon was determined using the Walkley and Black method as previously described (Rayment and Lyons, 2011).

### 2.3. Soil total metal and available metal analysis

Total soil metal (As, Ag, Cd, Cu, Co, Cr, Cu, Mn, Mo, Ni, Pb, Pd, Sb and Zn) analysis using acid digestion was carried out as previously described (De Silva et al., 2016).

Calcium chloride (CaCl<sub>2</sub>) was used to determine extractable metals. Air dried soil (10 g) was placed into a 250 ml polyethylene bottle with 0.01 M CaCl<sub>2</sub> at 20 °C in a 1:10 extraction ratio (W/V) and the suspension was shaken inversely for 2 h at 20 °C (Houba et al., 2000). Following mixing, a 60 ml aliquot of the suspension was decanted into a 100 ml centrifuge tube and centrifuged for 10 min at 1800 rpm. A 10 ml aliquot of the supernatant was then filtered using a syringe filter (0.45 µm), acidified with 4% HNO<sub>3</sub> (1:2



**Fig. 1.** Representative map of the study area within greater Melbourne, Victoria, Australia showing the major road network and the distribution of sampling sites. The sites included 27 roadside soil sites and 3 control sites. C = control soil samples, N = new roads (1.5–3 years of age), M = medium aged roads (5–13 years of age) and O = old roads ( $\geq 15$  years of age).

ratio) and analyzed for available metals (As, Ag, Cd, Cu, Co, Cr, Cu, Mn, Mo, Ni, Pb, Pd, Sb and Zn) using inductively coupled plasma mass spectrometry (ICPMS).

Reagent blanks representing 10% of the total sample population replicates and three certified reference materials (DO82–540, ERA) were incorporated into the analysis. The analytical results showed no signs of contamination and the accuracy was found to be consistently within 10% of the certified values.

#### 2.4. Analysis of polycyclic aromatic hydrocarbons (PAHs) and total petroleum hydrocarbons (TPH) in soil samples

Roadside soil hydrocarbons were analyzed using the method previously described (Haleyur et al., 2016). Hydrocarbons were measured using 5 g homogenated air dry soil and 15 ml hexane/acetone (Sigma Aldrich NSW, Australia) followed by 20 min of sonication. This was repeated 3 times (Haleyur et al., 2016). The extracts were centrifuged at 4000 rpm for 5 min. The samples were transferred to acetonitrile and the volume reduced to 5 ml total volume with a rotary evaporator as described in US-EPA 8310 method (Mansur et al., 2014). Hydrocarbons present in the soil were measured as previously described (Haleyur, 2016; Mansur et al., 2014). The resultant supernatant was transferred to a glass

vial (2 ml) and analyzed using gas chromatography-mass spectrometry (GC-MS). An Agilent 6890 GC was used with a capillary column of length 30 m, diameter 250  $\mu\text{m}$ , film thickness 0.25  $\mu\text{m}$ . Helium was used as a carrier gas, with a column temperature of 325  $^{\circ}\text{C}$  (Mansur et al., 2014). Calibration standards were prepared by diluting PAHs standards in hexane. A procedural hexane blank solution was passed through the columns with the same protocol as followed for soil extracts.

#### 2.5. Soil microbial basal respiration

Soil microbial basal respiration was measured as previously described (Dube et al., 2009) with modifications as detailed below. From each roadside soil (3 replicates), 25 g aliquots were sieved ( $< 2$  mm) and moistened to 60% MWHC (maximum water holding capacity). Soils were placed inside incubation jars together with 10 ml of 1 M NaOH. A blank (3 replicates), with no soil in the incubation jar was routinely included. The incubation jars were sealed with Parafilm®, covered with aluminum foil and left in an incubation chamber at  $22 \pm 1$   $^{\circ}\text{C}$  in the dark for 9 days. Every 3 days, the NaOH in the assay tubes were replaced with fresh solution. The  $\text{CO}_2$  present in the removed NaOH was titrated [after addition of  $\text{BaCl}_2$  (2 ml, 1.5 M)] using HCl (1 M) and phenolphthalein as the

indicator. The average HCl was used for the calculation. Carbon dioxide emitted was expressed as CO<sub>2</sub>/mg of dry soil/day using the following formula:

$$\text{CO}_2 \text{ evolved from the soil} = \text{mg of C or CO}_2 = (V-B) \text{ N.E}$$

Where:

B = volume of HCl (ml) from the titration of the blank (with no soil).

V = volume of HCl (ml) from the titration of the sample (with soil).

N = Normality of the acid for equivalent weight.

To express CO<sub>2</sub> evolved from the soil as mg of C use E = 6 or.

To express CO<sub>2</sub> evolved from the soil as mg of CO<sub>2</sub> use E = 22.

## 2.6. Microbial characterisation

Genomic DNA was extracted from roadside soil sample (0.25 g of soil) using a Power soil® DNA Isolation kit (MO BIO Laboratories, Inc USA) following the manufacturer's instructions. The purity of extracted DNA was checked using a Nanodrop-1000 spectrophotometer. All samples contained 12–39.4 ng DNA/μg.

The extracted DNA was subjected to polymerase chain reaction (PCR) in a Bio-RAD T100™ Thermo cycler. The roadside soil bacterial community was evaluated by polymerase chain reaction (PCR) using universal primers 907 F GC and 314 R on 16 S rDNA (Schäfer and Muyzer, 2001). Each PCR mixture contained DNA extract (2 μl), forward primer (2 μl, 10 mM), Go Taq Flexi buffer (1.0 μl, 5 X), MgCl<sub>2</sub> (5 μl, 25 mM), dNTPs mixture (1 μl, 10 mM), Taq polymerase enzyme (0.25 μl, 5 U/μl), and sterile nuclease-free water (27 μl). The amplification cycle was performed starting with 5 min at 95 °C, 30 min cycles of denaturation at 94 °C for 1 min, 52 °C for 1 min, 72 °C for 1 min, and final primer extension at 72 °C for 10 min. The amplified PCR products were checked by agarose gel electrophoresis (1.5% w/v). Following successful amplification, samples from 28 sites were suitable for DGGE analysis.

The soil fungal community was determined using nested polymerase chain reaction (PCR) amplification using internal transcribed spacer (ITS) region based primers, ITS1F, ITS2, ITS4 and ITS1F GC (Anderson and Parkin, 2007). Based on the PCR products amplification of the expected size band was observed in 20 samples.

DGGE was performed for selected soil genomic samples as described previously (Muyzer et al., 1993). The products of PCR were loaded onto a polyacrylamide gel (6% w/v, 1 mm thickness) in Tris-Acetate-EDTA (TAE) buffer. Gels (40–65% denaturing gradient) were run at 60 V for 18 h at 60 °C, stained using a silver staining protocol (Girvan et al., 2003), scanned (Epson V700) and saved as a TIFF file. Phoretix ID software was used to analyse the gel to generate dendrograms.

## 2.7. Statistical analysis

Pollution load index by metals for each site (new, medium and old roads) was evaluated using the following equation developed by Tomlinson et al. (1980).

$$\text{PLI} = (\text{CF}_1 \times \text{CF}_2 \times \text{CF}_3 \times \dots \times \text{CF}_n)^{1/n} \text{ (Godwin et al., 2015).}$$

Where: n = number of metals.

Contamination factor (CF) = Metal concentration in sediment/Background concentration of the metal. Control site metal concentrations were used as the background concentrations.

PLI > 1 is considered as polluted (Chakravarty and Patgiri, 2009).

Images of DGGE gels were scanned and subjected to analysis with Phoretix 1D software to create a dendrogram. The Phoretix 1D software will detect the presence and absence of the band and the intensity of the band. Based on the intensity, bands were manually selected. Unweighted pair group method with mathematical averages (UPGMA) was used to express microbial community similarity clusters. The Shannon index (H') was used to calculate the microbial diversity from DGGE profiles as described previously (Girvan et al., 2003) using the following formula:

$$H' = -\sum p_i \ln p_i$$

Where: H' = The Shannon index.

P<sub>i</sub> = intensity of the band i/total intensity of all bands in the lane.

Descriptive statistics were undertaken using SPSS Statistics (IBM version 21, 2012). Values are presented as means ± standard error. Roadside soil site means with control sites were compared using one-way ANOVA, significance was taken as P < 0.05. In situations where the data did not satisfy the Levene's test criteria (P < 0.05) the data were log ten or, square root transformed and retested for normality. In cases where the Levene's test criterion was still not met after transformation nonparametric tests were carried out (P < 0.05). Correlations between the roadside soil bacterial and fungal community and soil metal concentrations, PAHs, TPHs and soil physicochemical parameters were analyzed using 2-tailed Pearson correlation tests (significance P < 0.05), and principal component analysis (PCA) using XLSTAT 2015. Roadside soil physiochemical properties nutrients, soil texture, CaCl<sub>2</sub> extractable metals, total metals, PAHs, TPH, fungal and bacterial respiration/biomass and diversity index were incorporated into the principal component analysis.

## 3. Results

### 3.1. Physiochemical properties, nutrient status, PAHs and TPHs of roadside topsoil

Analysis of the physico-chemical properties of roadside soils studied found that the soils were generally neutral (mean pH of 6.9 ± 0.2), with mean soil concentrations of S, 15.5 ± 1.4 mg/kg; N, 40.5 ± 5.7 mg/kg and P, 42.6 ± 7.2 mg/kg. The soils studied were composed of sand 39 ± 3%; clay 27 ± 2.8% and silt 33 ± 3% (Table 1).

In terms of vehicle-related contaminants, the soil total petroleum hydrocarbon (TPH) concentration was similar in all soil samples studied (approximately 420 mg/kg, Table 1). However, roadside soil PAH concentrations varied considerably between sites (54–1555 mg/kg) (Table 1) and were significantly higher adjacent to older roads compared to medium aged roads (P = 0.048). PAH concentrations exceeded the Human Health Investigation Levels for detached residential houses (HIL-A = 20 mg/kg) in all studied soils (NEPC 2011).

Arsenic, Ag, Cd, Co, Cr, Cu, Mn, Mo, Ni, Pb, Pd, Sb and Zn were detected in roadside soils. Cadmium, Cr, Cu, Mn, Mo, Ni, Pb and Zn were found to be significantly accumulated in roadside soils compared to control soils (Table 2). Significantly higher concentrations of Cd (0.20 ± 0.03 mg/kg), Cr (23.5 ± 1.1 mg/kg), Cu (7.81 ± 0.74 mg/kg), Mn (258 ± 33 mg/kg), Ni (13.29 ± 1.15 mg/kg), Pb (45.3 ± 11.3 mg/kg) and Zn (32.5 ± 6.0 mg/kg) were found in roadside soils compared to control soils (Table 2). Manganese was the only metal in the current study to exceed the soil Ecological Investigation Level for Australia of 500 mg/kg (NEPM, 2013). Irrespective of the accumulated metal concentrations recorded from the roadside soils, CaCl<sub>2</sub> extractable metals were very low e.g. Cd (0.03 ± 0.01 mg/kg), Cu (0.08 ± 0.02 mg/kg), Ni (0.09 ± 0.01 mg/kg) and Zn (0.12 ± 0.038 mg/kg) (Table 2).



**Table 1**

Physicochemical, nutrient status, PAHs and TPHs analysis of the roadside soils sampled (n = 27) and non-roadside control soils (n = 3) from western Melbourne, Australia.

| Parameter            | Control soil |         |                 | Roadside soil |         |                   |
|----------------------|--------------|---------|-----------------|---------------|---------|-------------------|
|                      | Minimum      | Maximum | Mean $\pm$ SE   | Minimum       | Maximum | Mean $\pm$ SE     |
| pH                   | 5.2          | 5.2     | 5.19 $\pm$ 0.02 | 5.2           | 8.8     | 6.9 $\pm$ 0.2*    |
| EC ( $\mu$ S/cm)     | 42.5         | 56.4    | 48.8 $\pm$ 4.05 | <0.1          | 306.0   | 169.6 $\pm$ 14.6* |
| Moisture (%)         | 1.7          | 5.7     | 3.54 $\pm$ 1.1  | 0.1           | 31.2    | 17.0 $\pm$ 1.9*   |
| CEC (cmol (+)/kg)    | 5.8          | 12.5    | 8.81 $\pm$ 1.9  | 5.9           | 40.8    | 24.0 $\pm$ 1.7    |
| Sand (%)             | 25.9         | 42.1    | 36.2 $\pm$ 5.2  | 16.0          | 71.0    | 39.0 $\pm$ 3.0    |
| Clay (%)             | 39.2         | 71.8    | 56.12 $\pm$ 9.4 | 5.0           | 72.0    | 27.0 $\pm$ 3.0    |
| Silt (%)             | 2.4          | 3.5     | 3.15 $\pm$ 0.3  | 0.4           | 53.0    | 33.0 $\pm$ 3.0    |
| N (mg/kg)            | 11.2         | 89.2    | 42.1 $\pm$ 23.9 | 8.5           | 103.2   | 40.5 $\pm$ 5.7    |
| S (mg/kg)            | 8.4          | 11.9    | 9.7 $\pm$ 1.1   | 7.9           | 39.1    | 15.5 $\pm$ 1.4    |
| P (mg/kg)            | 14.4         | 31.8    | 20.3 $\pm$ 5.7  | 6.9           | 148.9   | 42.6 $\pm$ 7.2    |
| TOC (C %/g dry soil) | 0.5          | 2.8     | 1.9 $\pm$ 0.7   | 0.5           | 4.8     | 2.1 $\pm$ 0.2     |
| PAH (mg/kg)          | 59.3         | 202.3   | 140.1 $\pm$ 2.2 | 54.0          | 1555.0  | 250 $\pm$ 58.0    |
| TPH (mg/kg)          | 420.0        | 424.0   | 421.7 $\pm$ 1.1 | 417.0         | 438.4   | 424 $\pm$ 1.2     |

SE = standard error, EC = electrical conductivity, CEC = cation exchange capacity, N = nitrogen, P = phosphorus, S = sulfur, TOC = total organic carbon, PAH = petroleum aromatic hydrocarbon, TPH = total petroleum hydrocarbon, \* = significant difference in roadside soil compared to the control sites ( $P < 0.05$ ).

**Table 2**

Concentrations of soil metals (total and  $\text{CaCl}_2$  extractable) detected in roadside soils (n = 27) and non-roadside control soils (n = 3) of western Melbourne, Australia. Values are mean  $\pm$  standard error.

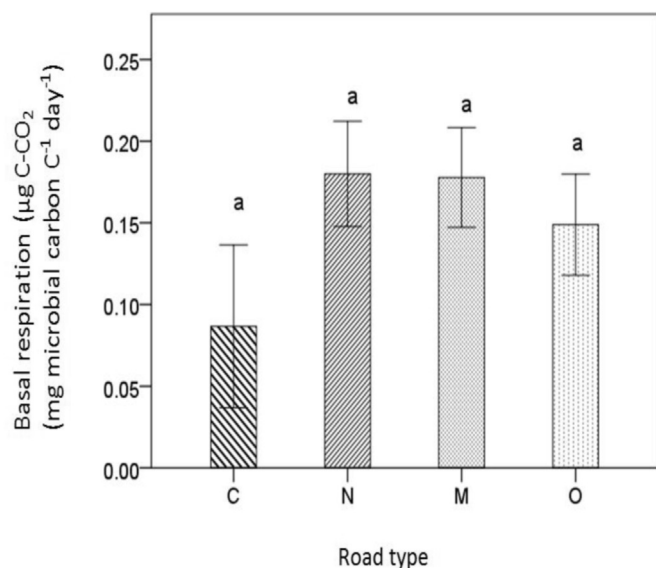
| Metal | Total metals mg/kg |         | Mean $\pm$ SE                       | $\text{CaCl}_2$ extractable metals mg/kg |         |                     |
|-------|--------------------|---------|-------------------------------------|------------------------------------------|---------|---------------------|
|       | Minimum            | Maximum |                                     | Minimum                                  | Maximum | Mean $\pm$ SE       |
| As C  | 1.94               | 11.59   | 6.79 $\pm$ 2.78                     | 0.04                                     | 0.24    | 0.114 $\pm$ 0.06    |
| As R  | <0.01              | 11.6    | 2.43 $\pm$ 0.45                     | <0.01                                    | 0.07    | 0.01 $\pm$ <0.01    |
| Ag C  | 0.09               | 0.36    | 0.22 $\pm$ 0.07                     | 0.01                                     | 0.32    | 0.18 $\pm$ 0.09     |
| Ag R  | <0.01              | 1.5     | 0.33 $\pm$ 0.07                     | 0.01                                     | 0.03    | 0.02 $\pm$ <0.01    |
| Cd C  | 0.02               | 0.11    | 0.054 $\pm$ 0.02                    | 0.002                                    | 0.023   | 0.01 $\pm$ 0.01     |
| Cd R  | 0.02               | 0.8     | <b>0.20 <math>\pm</math> 0.03*</b>  | 0.01                                     | 0.28    | 0.03 $\pm$ 0.01     |
| Co C  | 0.98               | 2.23    | 1.4 $\pm$ 0.41                      | 0.018                                    | 0.053   | 0.03 $\pm$ 0.01     |
| Co R  | 1                  | 18.2    | 6.1 $\pm$ 0.6                       | 0.01                                     | 0.16    | 0.03 $\pm$ 0.01     |
| Cr C  | 18.12              | 18.86   | 18.6 $\pm$ 0.23                     | n.d                                      | n.d     | n.d                 |
| Cr R  | 10.7               | 34      | <b>23.5 <math>\pm</math> 1.1*</b>   | n.d.                                     | n.d.    | n.d.                |
| Cu C  | 3.02               | 4.69    | 3.94 $\pm$ 0.49                     | 0.03                                     | 0.01    | 0.07 $\pm$ 0.02     |
| Cu R  | 3.02               | 23      | <b>7.81 <math>\pm</math> 0.74*</b>  | 0.02                                     | 0.41    | 0.08 $\pm$ 0.02     |
| Mn C  | 30.45              | 68.07   | 54.5 $\pm$ 12.1                     | 0.003                                    | 7       | 0.028 $\pm$ 0.02    |
| Mn R  | 92                 | 599     | <b>258 <math>\pm</math> 33*</b>     | 0.01                                     | 0.02    | 0.01 $\pm$ <0.01    |
| Mo C  | 3.55               | 6.74    | 4.65 $\pm$ 1.04                     | 0.04                                     | 0.12    | 0.08 $\pm$ 0.02     |
| Mo R  | 1.6                | 23.3    | 6.44 $\pm$ 0.93                     | 0                                        | 0.38    | 0.05 $\pm$ 0.02     |
| Ni C  | 30.45              | 68.07   | 54.51 $\pm$ 12.05                   | 0.003                                    | 0.07    | 0.03 $\pm$ 0.02     |
| Ni R  | 4.3                | 20.7    | <b>13.29 <math>\pm</math> 0.15*</b> | 0.02                                     | 0.24    | 0.09 $\pm$ 0.01     |
| Pb C  | 10.18              | 23.45   | 16.84 $\pm$ 3.8                     | 0.09                                     | 0.21    | 0.136 $\pm$ 0.04    |
| Pb R  | 16.2               | 144.6   | <b>45.3 <math>\pm</math> 11.3*</b>  | 0.01                                     | 0.18    | 0.04 $\pm$ 0.01     |
| Pd C  | 0.02               | 0.06    | 0.038 $\pm$ 0.02                    | n.d                                      | n.d     | n.d                 |
| Pd R  | 0.03               | 0.08    | 0.05 $\pm$ 0.00                     | 0.01                                     | 0.07    | 0.05 $\pm$ <0.05    |
| Sb C  | 0.19               | 6.95    | 2.72 $\pm$ 2.1                      | 0.05                                     | 0.27    | 0.18 $\pm$ 0.06     |
| Sb R  | 0.01               | 2.6     | 0.64 $\pm$ 0.13                     | 0.01                                     | 0.02    | 0.00 $\pm$ 0.00     |
| Zn C  | 5.39               | 11.59   | 8.41 $\pm$ 1.79                     | 0.09                                     | 0.042   | 0.023.37 $\pm$ 0.01 |
| Zn R  | 10.4               | 88.8    | <b>32.5 <math>\pm</math> 6.0*</b>   | 0.3                                      | 0.73    | 0.12 $\pm$ 0.038    |

n.d = not determined, C = control soil, R = roadside soil, \* = significant accumulation in roadside soil compared to the control sites ( $P < 0.05$ ).

The Pollution Load Index (PLI) for metals was calculated for new, medium and old aged roads. All three road types showed evidence of pollution with metals as per the  $\text{PLI} > 1$  (Chakravarty and Patgiri, 2009). The highest PLI was calculated was for old roads  $\text{PLI} = 2.23$  followed by medium aged roads  $\text{PLI} = 1.74$  and new roads  $\text{PLI} = 1.58$  were the least polluted.

### 3.2. Soil respiration

Soil respiration rates did not vary significantly among the different roadside or control soils (Fig. 2). Metals that were significantly accumulated in roadside soils compared to controls (Cd, Cr, Cu, Mn, Ni, Pb and Zn) did not show a significant correlation with



**Fig. 2.** Roadside soil basal respiration [ $\mu\text{g C-CO}_2$  (mg microbial carbon  $\text{C}^{-1}$  day $^{-1}$ )]. Bars represents C = control soil samples, N = new aged roads (1–3 years of age), M = medium aged roads (5–13 years of age) and O = old roads ( $\geq 15$  years of age),  $n = 3$  for control soils and  $n = 9$  for each road age, lower case letters on the bar represents significance, error bar represents  $\pm$  standard error.

roadside soil respiration (Table 3). In addition, neither soil texture, soil nutrients nor available metals in soil showed any significant correlation with soil basal respiration (Table 3) and microbial diversity.

### 3.3. Bacterial diversity

Denaturing gradient gel electrophoresis (DGGE) analysis of roadside soil bacterial diversity confirmed a significant increase in soil bacterial diversity in soils adjacent to new roads ( $P = 0.016$ ), medium-aged roads ( $P = 0.044$ ) and old roads ( $P = 0.015$ ) compared to control soil bacterial diversity (Fig. 3a). However, there was little variation observed in bacterial diversity among soils collected adjacent to different aged roads; irrespective of the exposure time to vehicular emissions the roadside soil bacterial diversity distribution was the same across all categories of road age. Bacterial diversity was significantly correlated with soil EC ( $r = 0.443$ ,  $P = 0.015$ ), and was significantly reduced as the concentrations of As ( $r = -0.625$ ,  $P < 0.001$ ) and Sb ( $r = -0.732$ ,  $P < 0.001$ ) increased in roadside soil (Table 3).

### 3.4. Fungal diversity

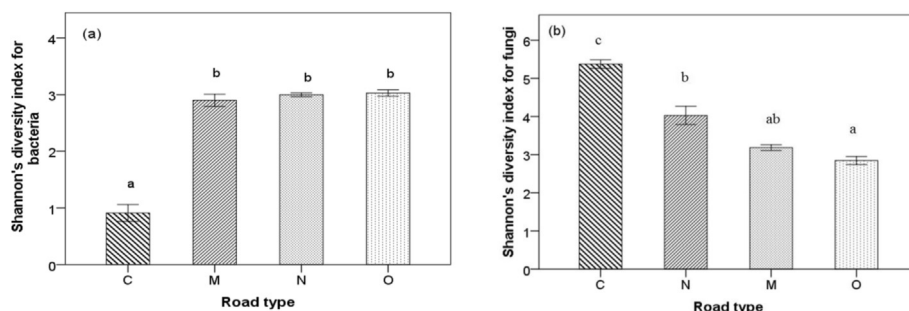
Control sites were significantly higher in fungal diversity compared to the fungal diversity of soil collected adjacent to new roads ( $P = 0.035$ ), medium-aged ( $P = 0.023$ ) and old roads ( $P = 0.004$ ) (Fig. 3b). Fungal diversity was significantly higher in soils adjacent to new roads and declined as the age of the roads increased, i.e. the fungal diversity adjacent to new roads was significantly different ( $P = 0.001$ ) to that found in old roadside soils (Fig. 3b). Non-parametric test analysis in SPSS for independent samples showed a significant relationship ( $P = 0.002$ ) between road age and roadside soil fungal diversity i.e. with increased age of the road, the fungal diversity decreased. Increased soil nitrogen concentration was correlated with increased fungal diversity ( $r = 0.399$ ,  $P = 0.037$ ). In addition, fungal diversity in roadside soil was negatively correlated with the presence of Cu ( $r = -0.462$ ,  $P = 0.01$ ), Pb ( $r = -0.412$ ,  $P = 0.024$ ) and Zn ( $r = -0.439$ ,  $P = 0.017$ ).

**Table 3**

Pearson correlation coefficients ( $r$ ) of microbial diversity with soil physico-chemical properties, nutrients, total metals and hydrocarbons.

| Properties           |                                   | Microbial diversity |          |
|----------------------|-----------------------------------|---------------------|----------|
|                      |                                   | Bacteria            | Fungi    |
| Physical properties  | Clay (%)                          | −0.157              | 0.153    |
|                      | Sand (%)                          | 0.17                | −0.235   |
|                      | Silt (%)                          | −0.172              | 0.292    |
|                      | pH                                | 0.349               | 0.247    |
|                      | CEC (meq/100 g)                   | 0.315               | 0.151    |
|                      | EC ( $\mu\text{S}/\text{cm}$ )    | 0.443*              | −0.160   |
|                      | Moisture (%)                      | 0.245               | −0.017   |
|                      | N ( $\text{NO}_2 + \text{NO}_3$ ) | −0.309              | 0.399*   |
| Nutrients (mg/kg)    | P ( $\text{PO}_4^{3-}$ )          | 0.062               | −0.276   |
|                      | S ( $\text{SO}_4^{2-}$ )          | 0.196               | −0.237   |
|                      | C (TOC)                           | 0.153               | −0.351   |
|                      | As                                | −0.625**            | 0.125    |
| Total metals (mg/kg) | Cd                                | 0.224               | −0.124   |
|                      | Cr                                | 0.198               | −0.108   |
|                      | Cu                                | 0.272               | −0.462** |
|                      | Mn                                | 0.256               | 0.099    |
|                      | Ni                                | 0.331               | −0.285   |
|                      | Pb                                | 0.166               | −0.412*  |
|                      | Pd                                | −0.036              | 0.485**  |
|                      | Sb                                | −0.732**            | 0.161    |
|                      | Zn                                | 0.225               | −0.439*  |
|                      | TPH                               | 0.123               | −0.250   |
|                      | PAH                               | 0.121               | −0.271   |
|                      | BD                                | 1                   | −0.489** |
| Microbial diversity  | FD                                | −0.489**            | 1        |

BD = bacterial diversity, BR = basal respiration, CEC = cation exchange capacity, EC = electrical conductivity, FD = fungal diversity, TOC = total organic carbon, PAHs = petroleum aromatic hydrocarbons, TPH = total petroleum hydrocarbons, \* $P < 0.05$ , \*\* $P < 0.01$ .



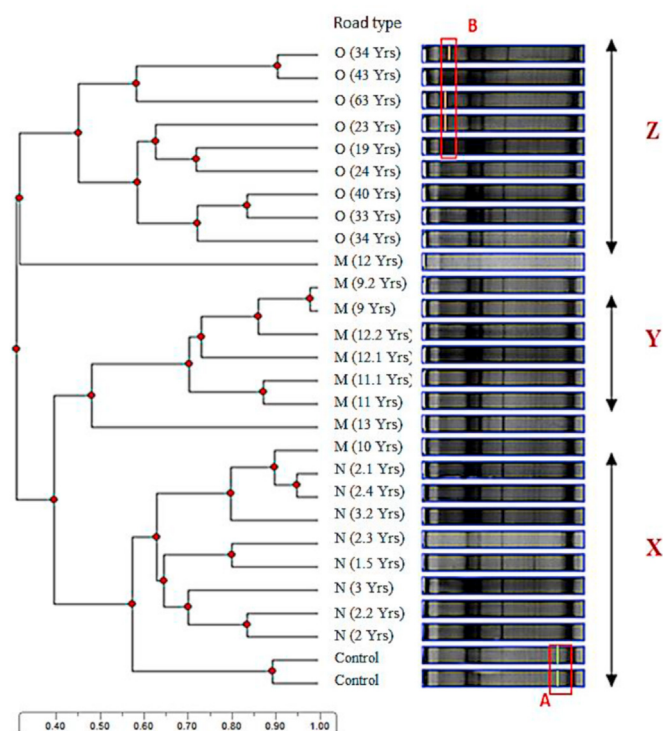
**Fig. 3.** The Shannon Diversity Index (mean ± standard error) for (a) bacteria and (b) fungi in roadside soil. Bars represent new (1–4 years of age), medium (5–13 years of age), and old (≥15 years of age) roads.

(Table 3). Those metals were also found to be significantly accumulated in Melbourne roadside soils compared to control soils (Table 2).

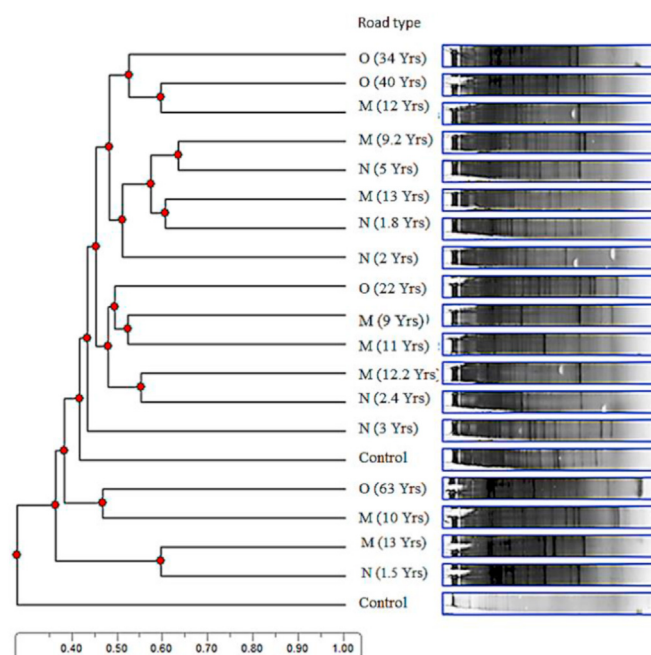
### 3.5. Soil microbial community fingerprinting

Polymerase chain reaction–DGGE (PCR–DGGE) analysis of the bacterial and fungal community of the soils is shown in Figs. 4 and 5 respectively. Many common bands were detected among the soils suggesting that the presence of many bacterial taxa was common to all soils tested (Fig. 4).

Analysis of the community fingerprint from each soil identified three distinct clusters, X, Y and Z (Fig. 4). The first cluster (X) consisted of the bacterial profiles from soil from the control site and soils exposed to vehicular emissions for a short period of time (2–3 years); the second cluster (Y) consisted of soil collected from beside



**Fig. 4.** Unweighted Pair Group Method with Arithmetic Mean (UPGMA) dendrogram of bacterial community (based on 16 S rDNA denaturing gradient gel electrophoresis (DGGE) profiles of some roadside soil samples. X = recent exposure, Y = moderate exposure, and Z = long term exposure to vehicular emissions, A and B represents microbial bands.



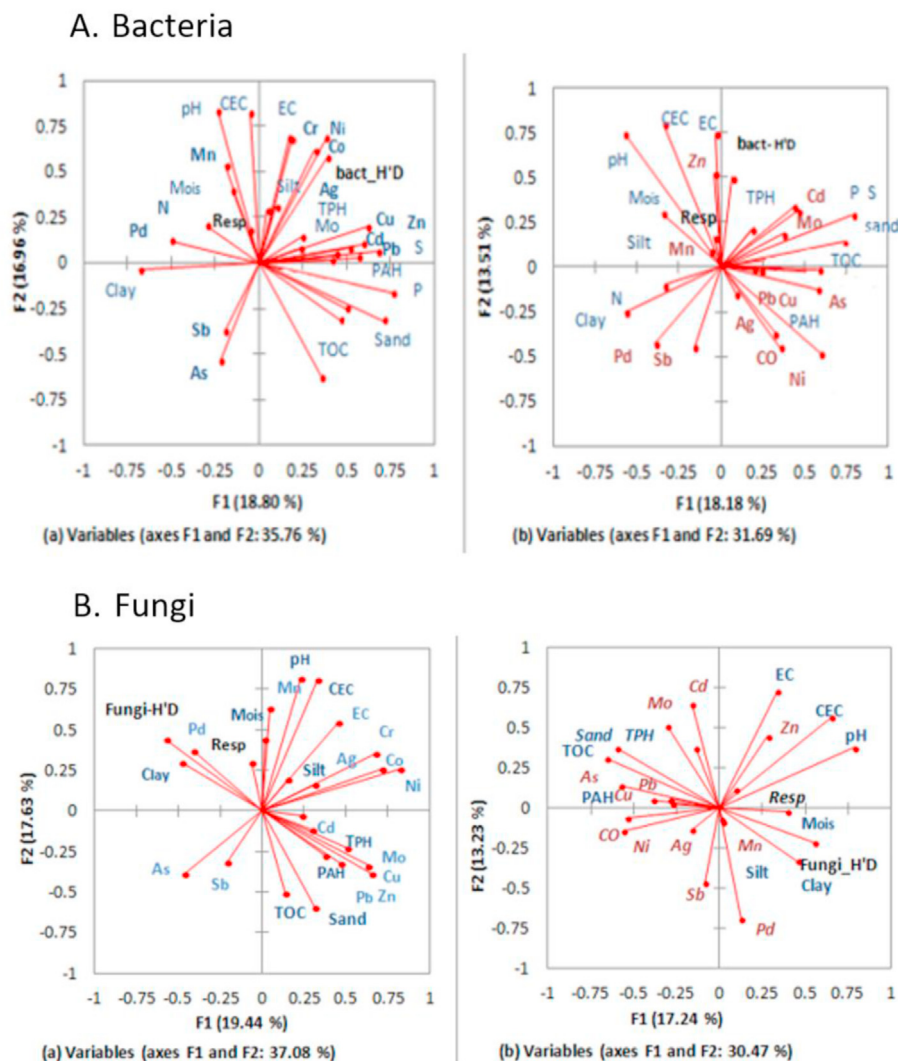
**Fig. 5.** Unweighted Pair Group Method with Arithmetic Mean (UPGMA) dendrogram of fungal community denaturing gradient gel electrophoresis (DGGE) profiles of the roadside soil samples for soil with DNA amplification.

medium-aged roads (5–12 years) while the third cluster (Z) consisted of soil collected from beside older roads (>15 years) (Fig. 4). In addition, some bands appearing in the control soil (bands shown as A in Fig. 4) were absent in other roadside soil sites, while some bands unique to roadside soil exposed to vehicular emissions for a long period of time (bands shown as B in Fig. 4) were absent in control sites and other roadside soil.

Denaturing gradient gel electrophoresis analysis of the fungal community also identified many common bands across all sites (Fig. 5). No distinct clustering was identified in fungal DGGE dendrogram analysis.

### 3.6. Soil microbes and roadside soil pollution

Principal component analysis was used to assess the impact of metals and PAHs on the roadside soil microbial community (Fig. 6). The first and second principal components for bacteria were able to explain 35.8% of the variation between the variables and included total metals, TPH, PAHs and soil physical properties (Fig. 6A–a) while a further 31.7% of the variation was explained by available



**Fig. 6.** Principal Component Analysis (PCA1 based on biological and physio-chemical properties of the soils; (A) Bacteria and (B) Fungi H'D are Shannon Diversity indices of microbial diversities from PCR-DGGE. (a) Represents total soil metals and soil physical properties (b) represents the PCA analysis for soil available metals and other soil properties studied.

metals, TPH, PAHs and soil physical properties (Fig. 6A-b). Bacterial diversity was negatively influenced by the presence of total metals of Sb and As and available Pd and Sb in soil (Fig. 6A-b). The presence of clay also showed a negative effect on roadside soil bacterial diversity (Fig. 6A-a).

The first and second principal components for fungi was able to explain 37.1% of the variance between the variables including fungal diversity, soil total metals, TPH, PAHs and soil physical properties (Fig. 6B-a) while 30.5% was explained by the soil available metals, TPH, PAHs and other soil properties (Fig. 6B-b). Fungal diversity was associated with soil physical properties including clay and moisture content, and was negatively influenced by the TPH, PAHs, TOC and metals (both for total and available) of Cd, Cu, Mo and Pb (Fig. 6B); available As (Fig. 6B-b) and total Zn (Fig. 6B-a) concentrations in roadside soil.

## 4. Discussion

### 4.1. Roadside soil pollution and microbial populations

The findings of the current study showed that the basal

respiration/microbial biomass was the same among roadside soil and control soils; however, an increase in microbial basal respiration/biomass in roadside soil compared control soil was also noted (Fig. 2). Microbial respiration was closely related to the soil moisture content and the presence of some metals in roadside soil (Fig. 6). Low moisture content in soil can inhibit microbial activity by lowering intracellular water potential and thus reducing hydration (Stark and Firestone, 1995). Therefore moisture content may be an important factor determining the microbial biomass in soil. Post and Beeby (1993) and Adelasoye and Ojo (2014) also found the microbial population to be higher closer to the roadside than further away. High metal concentrations are usually known to reduce microbial biomass, as elevated concentrations of metals in soils can decrease the population size of the microbial communities. However, in field aged soil the impacts of pollutants like metals on microbial biomass can be extremely diverse; i.e. 'moderate' metal concentrations have been previously reported to have no significant effect or have a mild stimulatory effect (Khan and Scullion, 2002; Máthé-Gáspár, 2005). To date the effects of pollutants on different microbial populations on roadside soil has not been well documented.



#### 4.2. Roadside soil pollution and microbial diversity

For any ecosystem, the resilience of the soil is closely associated with microbial biodiversity, such that increased microbial biodiversity of a soil increases soil resilience (Arias et al., 2005). A significant increase in bacterial diversity in roadside soil compared to control soil bacterial diversity was observed (Fig. 3). Roadside soil bacterial diversity was significantly higher (new roads:  $P = 0.016$ , medium-aged roads:  $P = 0.044$  and old roads:  $P = 0.015$ ) compared to the control soil bacterial diversity. According to principal component analysis, the addition of some metals to roadside soils like Cd, Co, Cr, Cu, Ni Pb and Zn from vehicular emission seems to be beneficial to the roadside soil microbes while some microbes were significantly reduced as the concentrations of As ( $r = -0.625$ ,  $P < 0.001$ ) and Sb ( $r = -0.732$ ,  $P < 0.001$ ) increased in roadside soil (Table 3). The presence of PAH-resistant bacteria has been reported in roadside soil (Thavamani et al., 2012; Marchand et al., 2017). Therefore the presence of such bacteria (e.g. PAH resistant bacteria) could increase the soil bacterial diversity in roadside soil. Soil PAH concentrations could be increased by both natural (e.g. bush fire) and anthropogenic (e.g. vehicular emissions) sources; all soils studied contained high concentrations of PAHs (Table 1). We expected a significant increase in roadside soil bacterial diversity; however, we also expected a significant effect of road age on roadside soil bacterial diversity. The absence of a road age effect on roadside soil bacterial diversity indicates that when roadside soil bacteria are exposed to contaminants  $\leq 1.5$  years (youngest age of the road studied was only 1.5 years old), they were able to acquire the required ability to survive and thrive in roadside soil or simply that the accumulated metals in roadside soil were not bioavailable.

In contrast to the high bacterial diversity in roadside soil, fungal diversity in roadside soil was significantly reduced (new roads  $P = 0.035$ , medium-aged roads  $P = 0.023$ , old roads  $P = 0.004$ ) compared to control soil (Fig. 3). A significant difference among new and old aged roadside soil fungal diversity ( $P = 0.001$ ) was found: i.e. new age roadside soil fungal diversity > old aged road fungal diversity. The metal effects on fungal diversity (Fig. 3) demonstrated an inverse relationship with roadside soil pollution level index (PLI). Fungi were less affected in reduced metal-contaminated sites: in soils close to a new aged road where the PLI (1.58) was lower fungal diversity was high; old roadside soils with higher PLI (2.25) exhibiting a high fungal diversity were more effected. As the fungal diversity in medium aged roadside soil was not significantly different from that found in soils close to new or old aged roads, this may indicate that the effect on the fungal community may be cumulative, taking more than 13 years. This age effect on the fungal community was also confirmed by the significant correlation between fungal diversity and age of the roads ( $P = 0.002$ ). According to principal component analyses, roadside metals like Cd, Cu, Mo, Pb and Zn together with the presence of PAH and TPH negatively impacted fungal diversity (Fig. 6B). In fact, the fungal diversity in roadside soil was negatively correlated with the presence of Cu ( $r = -0.462$ ,  $P = 0.01$ ), Pb ( $r = -0.412$ ,  $P = 0.024$ ) and Zn ( $r = -0.439$ ,  $P = 0.017$ ) (Table 3). Those metals were also found to be significantly accumulated in roadside soils compared to control soils (Table 2). Similar inhibitory effects on the soil fungal population have previously been shown for Cu and Zn enrichment in non-roadside soil (Klimek, 2012). These findings suggest that the fungal population within roadside soil may be compromised when exposed to roadside vehicular emitted contaminants compared to bacteria.

The presence of a negative correlation ( $r = -0.489$   $P = 0.007$ ) between roadside soil bacterial diversity and fungal diversity demonstrated a decrease in one community (the fungal community), corresponding to an increase in another (bacterial

community), perhaps with better survival ability (Atlas, 1984). Compared to fungi, bacteria may be better adapted to chronic toxicity or stress in long term contaminated soil due to their wide substrate utilization profile and high metabolic activity (Chen et al., 2014). Additionally, bacteria in roadside soils compete with fungi for substrates. This is the first time such contrasting behaviour of a bacterial and fungal community from roadside soil has been reported.

#### 4.3. Roadside soil pollution and microbial fingerprinting

To investigate the differences between the fungal and bacterial communities in roadside soil further, DGGE analysis was carried out. DGGE analysis classified the roadside soil bacteria into three different clusters: X (soil adjacent to new roads), Y (soil adjacent to medium-aged roads), and Z (soil adjacent to old roads) (Fig. 4). Together with this clustering effect, bacterial DGGE analysis identified the presence of many different bands in roadside soil. In contrast the fungal DGGE profile did not demonstrate any clustering effect or definite bands (Fig. 5). The sampling sites were at least 1 km apart from each other. If the clustering was due to location and proximity of the sampling sites, similar clusters should have been reported for both bacteria and fungal communities.

In the DGGE profile, the presence of a single band generally suggests the presence of a unique species (Wallace et al., 2002). Some of the bands found in the bacterial DGGE profile were only seen in control sites (e.g. band A, Fig. 4) and disappeared in soils collected from beside old roads, while some new bands appeared (e.g. band B, Fig. 4) in soils from beside old roads. The samples associated with long- and medium-term metal exposure were clearly separated from the control, suggesting that the accumulation of metals over a period resulted in changes in the roadside soil bacterial community. This changing microbial diversity may indicate that metal resistant species increased within the population allowing the bacterial community to continue to be active even as metals accumulated over time. We hypothesized that the roadside fungal community did not have a large number of metal-resistant species (in contrast to the bacterial community). This may explain the reduced diversity seen in the fungal population (Fig. 5). A potential decline in the fungal community in roadside soil would likely result in a significant impact in terms of soil quality.

The activity and community composition of soil microorganisms are closely related to soil fertility and environmental quality. The estimated affected land area for roadside soil in Australia itself is large, 80 million km<sup>2</sup>, and in the world far more (De Silva et al., 2020; Gao et al., 2019; Liu et al., 2020). Therefore these soils are likely to be disturbed in terms of biogeochemical processes and may affect the normal functionality of soil ecosystem services.

### 5. Conclusions

The concentrations of metals, PAHs and TPH in roadside soil has the potential to impact microbial activity and function. Our results suggest that vehicular emitted metals increased roadside soil microbial respiration, reduced fungal diversity, and increased bacterial diversity. Fungal diversity appeared to be suppressed by the presence of metals, TPH and PAHs in roadside soil where the bacterial diversity was not affected. Interestingly, three clusters of bacteria communities were found, representing roadside soil exposed to contaminants over different periods of time. For the fungal community, no age effect of contaminants was found. Thus, as per our original hypothesis, exposure to long-term metal pollution from vehicle emissions affected the diversity and activity of the roadside soil microbial community with different responses from fungal communities compared to bacterial communities. Our

study is the first to directly compare roadside bacterial and fungal communities and document different impacts of roadside soil contaminants on bacterial and fungal communities.

## Author contributions

Conceptualization, S.D.S., A.S.B. and S.M.R.; methodology, S.D.S., A.S.B., E.S. and S.M.R.; validation, S.D.S., A.S.B., E.S. and S.M.R.; formal analysis, S.D.S., and E.S.; investigation, S.D.S., A.S.B., E.S., D.V.I. and S.M.R.; data curation, S.D.S., A.S.B., E.S., D.V.I. and S.M.R.; writing—original draft preparation, S.D.S.; writing—review and editing, S.D.S., A.S.B., E.S., D.V.I. and S.M.R.; visualization, S.D.S.; supervision, A.S.B. and S.M.R.; project administration, S.D.S. All authors have read and agreed to the published version of the manuscript.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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